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Notifications 

3. Solve homogeneous system

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Recitation due Jan 24, 2024 20:30 IST

Solving homogeneous equations with MATLAB

(External resource)

(1.0 points possible)

Backslash and homogeneous equations

Previously, we saw that to solve a linear system $Ax = b$ for x using MATLAB, one uses the

```
x = A\b;
```

Let's see what happens when we use this approach for the homogeneous equation $Ax = 0$

```
A = [ -0.7950    -2.3851   -0.9578    0.1628   -0.1628;
      -0.1236   -0.3708   -0.8113    0.6877   -0.6877;
      -0.5517   -1.6551    0.5935   -1.1452    1.1452;
           0.3406    1.0219   -0.1154    0.4560   -0.4560];
```

Solve for

```
x
```

and then find the dimension of the nullspace, in a variable called

```
dim
```

To do this, it may be helpful to use the function `zeros( )`, which creates an $m \times n$ matrix

```
zeros(m,n);
```

Script ?

 Save  Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 warning off backtrace
2 % The matrix A below
3 A = [ -0.7950    -2.3851   -0.9578    0.1628   -0.1628;
4       -0.1236   -0.3708   -0.8113    0.6877   -0.6877;
5       -0.5517   -1.6551    0.5935   -1.1452    1.1452;
6           0.3406    1.0219   -0.1154    0.4560   -0.4560];
7 %Use the backslash command to solve for x
8 x = A \ zeros(size(A,1),1)
9
10 % Find the dimension of the nullspace to see if this makes any sense.
11 % Store the dimension in a variable called dim
```

Find a nonzero solution (External resource)

(1.0 points possible)

Find a nonzero solution

Find any nonzero solution $\mathbf{x} \neq \mathbf{0}$ to the homogeneous linear equation $\mathbf{Ax} = \mathbf{0}$, where th
the matrix in the proble above:

A = [-0.7950 -2.3851 -0.9578 0.1628 -0.1628;
 -0.1236 -0.3708 -0.8113 0.6877 -0.6877;
 -0.5517 -1.6551 0.5935 -1.1452 1.1452;
 0.3406 1.0219 -0.1154 0.4560 -0.4560];

Check you answer by creating a vector

b=A*x

Script ?

 Save

 Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

1 %Find a nonzero solution x to Ax=0.
2 A = [-0.7950 -2.3851 -0.9578 0.1628 -0.1628;
3 -0.1236 -0.3708 -0.8113 0.6877 -0.6877;
4 -0.5517 -1.6551 0.5935 -1.1452 1.1452;
5 0.3406 1.0219 -0.1154 0.4560 -0.4560];
6
7 N = null(A);
8 x = N(:,1);
9 %Check that x is a solution:
10 b = A*x

 Run Script



Assessment: All Tests Passed

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 Check that x is a non-zero solution

3. Solve homogeneous system

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